Major Contributors to This Report

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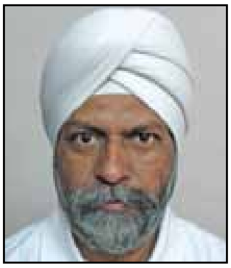

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• International UL 2043 standard is for fire test for heat and visible smoke release for discrete products installed in AHU space. It addresses the concern for contribution to smoke density or flame propagation by the equipment during a fire.

"Manufacturers follow the standards and guidelines prescribed in the handbook 'UVGI for Air and Surface disinfection' authored by Wladyslaw Kowalski. For buyers, it is important to check the intensity test certificate. These certificates provided by authorised laboratories to the manufacturer can be a proof for the amount of intensity used by a specific product for irradiation," opines Sreejith Swaminathan, Senior Product Manager, Zedbee Technologies.

Practices and precautions

While many people have started installing UVGI systems in public places, these require maintenance and monitoring with respect to the lamp used to discharge the UVC radiation. "UVGI is a science and use of UVGI technology must be understood by the user before applying, as it can be harmful to humans, and if not monitored properly, ineffective for the pathogens that it is required to destroy, with serious consequences of infections," says Rohatgi.

Unlike UVA and UVB, lower wavelengths of UVC prevent its penetration deep into the skin. Thus, there is not much risk when exposure is for a limited period of time. But, prolonged continuous exposure has been linked to reduced immunity and can cause cancer. NIOSH and IEC have fixed a maximum UVC of 254nm as allowable dose for human exposure in an eight hour workday. This should be pursued with a period of rest to allow the body to regenerate to withstand UVC again.

"LED based UVC products are preferred due to long lasting durability, and they are not affected by temperature. We have partnered with Council of Scientific & Industrial Research to work on UVC systems to sanitise aircrafts, and we have already taken up sanitisation projects in metros along with the DMRC (Delhi Metro Rail Corporation)," says Atul Agarwal, Technical Director, Reiz Electrocontrols.

Mercury lamp sources generate ozone that poses severe harm to living cells of the human body, leading to chest pain, coughing, throat irritation, and can even be fatal. To counter this issue, leading manufacturers use UVC tubes made of glass or quartz, which reduce the emission of ozone into the atmosphere.

To UVC or not to UVC?

After one year of the pandemic, CDC (USA) concluded after extensive studies that Covid-19 is primarily an airborne disease. In fact, one of the studies concluded that 9,900-plus cases out of 10,000 are airborne infection cases. Thus, the need to disinfect the air in enclosed spaces is very, very vital.

Here is how UVC solutions could be of help:

1. Install UVC solutions in central air-conditioning units to disinfect the air so that infection cannot spread from room to room.

2. Install in-duct UVC solutions to treat circulated air.

3. To prevent in-room infection, which is the most popular mode of infection, it is imperative to disinfect the air in the room, for which you have following options:

• *Enclosed UVC devices.* These suck in air and subject it to a very high dose of UVC energy to disinfect the air. This option, while effective for small rooms, does not disinfect fast enough in larger rooms to prevent infection.

• *Enclosed room air filters with UVC.* These often use HEPA filters, which limit air flow and have a slow air delivery, so are again not very effective to prevent infection in a room.

• *Upper-room air UVC solutions.* These have the fastest mode of room air disinfection. Studies have shown that even with basic air circulation, air disinfection rates equivalent to 24 air-changes-per-hour have been achieved.

Overall, it is clear that UVC does present many solutions to fight coronavirus. Be it in our homes or public spaces, there seem to be solutions we can invest in. For many businesses, these may be the best investment to ensure continuity of their operations by providing safety and security for their customers, and their employees. Upper-room luminaries and lower-room luminaries seem to stand out amongst all, for most cases. EFY

This report was compiled by Abbinaya Kuzhanthaivel with key inputs from Vineet Rohatgi, Director, Binay Opto Electronics, and Hina Gupta, Director, ASHRAE India.